

Course- B.Tech.	Type- Core
Course Code- 2025CSET106	Course Name- Discrete Mathematical Structures
Year- I	Semester- Even (2 nd Semester)

Tutorial-1

Tutorial No.	Name	CO1	CO 2	CO3
1	Relation: Graphical, Matrix Representation of relations, Types of relations	✓	--	✓

Objective:

1. Students will be able to represent relations as matrix and digraph.
2. Students will be able to recognize types of relations and functions.
3. Students will be able to understand the difference between relations and functions.

Questions

1. Represent the relation $R = \{(1,2), (1,4), (2,1), (2,3), (3,3), (4,1), (4,2)\}$ on the set $A = \{1, 2, 3, 4\}$ by matrix and Digraph.
2. Represent the divisibility relation on the set $A = \{1, 2, \dots, 6\}$ by a digraph.
3. Let R_1 and R_2 be given by the matrices:

$$R_1 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \text{ and } R_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

- a) Check reflexive, symmetric, antisymmetric and transitive properties.
 - b) Find R_1^c , $R_1 \cup R_2$, $R_1 \cap R_2$ and $R_1 \circ R_2$.
4. Let $R = \{(1,2), (2,2), (2,3), (5,4)\}$ be a relation on $S = \{1,2,3,4,5\}$. Obtain reflexive, symmetric and transitive closures of R .

5. Represent the relation

$$R = \{(1,1), (1,3), (2,2), (2,4), (3,1), (4,2)\}$$

on the set

$$A = \{1,2,3,4\}$$

by matrix.

“Set theory is the foundation upon which modern mathematics is built.”

— *Georg Cantor*